

An Innovative Industry Program in A New Era of Multimedia with Generative AI*

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Abstract

The ACM Multimedia 2024 industry program offers a unique platform for fostering collaboration between academia and industry. This year's program features a diverse range of industry keynotes, expert talks, seminars, and demonstrations, showcasing the latest advancements in multimedia technology. Renowned experts from industry and academia will share their insights on topics such as generative AI, automotive design, computer vision, spatial experience, healthcare, and more. Attendees will have the opportunity to network with industry leaders, learn about cutting-edge technologies, and explore potential collaborations. The industry program highlights the growing importance of multimedia technology in various domains and demonstrates the innovative ways in which AI and other emerging technologies are transforming industries. By participating in this program, attendees can gain valuable knowledge, expand their professional networks, and contribute to the advancement of the field.

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1 Introduction

The ACM Multimedia 2024 conference stands as a pivotal platform for fostering collaboration and innovation within the multimedia research community. This year, the industry program has been meticulously curated to offer a dynamic and engaging experience, bridging the gap between academia and industry.

By introducing industry keynotes, expert talks, and seminars, we aim to attract high-profile industry researchers and showcase their valuable insights and practical experiences. The program boasts a rich array of technical talks from senior and expert researchers representing a diverse range of industry companies, from tech giants to startups. This diverse representation ensures that the content is

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relevant, engaging, and informative. Key highlights are as follows:

(1) **Industry Keynotes:** Renowned experts will deliver keynote speeches, offering their unique perspectives on the future of multimedia technology. (2) **Industry Seminars:** A dedicated seminar will showcase the latest advancements in a particular area of multimedia technology. (3) **Industry Expert Talks:** Gain valuable insights from industry leaders who will delve into specific topics and share their practical experiences. (4) **Industry Demonstrations:** Witness firsthand the groundbreaking innovations shaping the multimedia landscape through interactive demonstrations.

The industry program offers a unique opportunity to network with industry leaders, learn about the latest trends, and contribute to the advancement of the field. By participating in this program, attendees can expand their professional networks, stay updated on the latest developments, and gain practical insights into real-world applications of multimedia technology.

We believe that this year's industry program is a significant step forward in fostering collaboration between academia and industry. By providing a platform for researchers and industry professionals to connect, share ideas, and explore new opportunities, we aim to drive innovation and advance the field of multimedia.

2 Industry Keynotes

Dr. Benoit Huet, a renowned expert in AI and Data Science, will deliver the keynote focusing on eyonis™LCS, a cutting-edge AI/ML-based SaMD designed to revolutionize lung cancer diagnostics. By accurately identifying malignant nodules from CT scans, eyonis™LCS can enable earlier diagnosis and potentially save millions of lives. Dr. Huet's extensive experience in AI and his leadership at MEDIAN Technologies make him uniquely qualified to share insights into the future of AI-powered healthcare. This keynote offers a unique opportunity to learn about the transformative power of AI in medical imaging, the latest advancements in lung cancer diagnosis, and the potential of AI to improve patient outcomes.

As the second industry keynote, Mr. Des Keane, Engineering Director of Video Infrastructure at YouTube, will explore audio and video infrastructure, analytics, and processing powered by AI technologies at scale. With over 20 years of experience at Google, Mr. Keane is a leader in large-scale video systems. This talk unveils how the latest AI technologies are transforming video experiences on YouTube and beyond.

3 Industry Seminars

NEC Seminar: R&D on Biometrics, Video Analytics, and Generative AI. This seminar will showcase NEC's latest advancements

in biometrics, video analytics, and generative AI. Dr. Jianquan Liu will discuss the convergence of video analytics and generative AI, highlighting industry applications and NEC's innovative narrative summarization framework. Dr. Masafumi Oyamada will explore NEC's research on sustainable AI agents, focusing on self-evolving LLMs/VLMs, RAG, and action models. Finally, Dr. Atsushi Ito will introduce NEC's multimodal authentication terminal, which combines face and iris recognition for enhanced security. This seminar offers valuable insights into the future of AI-driven technologies and their practical applications. This seminar is a must-attend for researchers, developers, and industry professionals interested in the latest advancements in AI and its applications.

4 Industry Expert Talks

The industry expert talks offer a fascinating glimpse into the future of multimedia technology, with a particular focus on generative AI and its applications across various domains. From automotive to healthcare, these talks highlight the innovative ways in which AI is transforming industries and addressing pressing societal challenges.

Takuya Nanri from Nissan Motor Co., Ltd., will share insights into the company's groundbreaking work in applying generative AI to automotive design. His talk explores how AI can streamline the product design process, accelerating development and improving efficiency. He will discuss the use of AI for generating and evaluating product designs, as well as for speeding up simulations.

Takayoshi Yamashita from Chubu University, offered a unique perspective, drawing from his extensive experience in both industry and academia. His talk will explore the evolution of computer vision research and development over the past two decades, providing valuable insights for researchers and practitioners alike.

Haohong Wang from TCL, a pioneer in AI filmmaking, will discuss how AI is transforming the industry. His talk will cover the latest advancements in AI-driven tools and their impact on filmmaking workflows, including storytelling, production, and visual quality. Don't miss this opportunity to learn about the future of AI in the creative arts.

Guan-Ming Su from Dolby Labs, will delve into the challenges and opportunities of spatial experience, a rapidly growing area of research. His talk will cover the latest advancements in 3D spatial representations and standardization efforts, providing valuable insights for developers and researchers.

Guoming Wang from Zhejiang Mingji Technology Co., Ltd., will focus on the application of multimodal large language models in ultrasound diagnosis. His research aims to address the challenges of data scarcity and model limitations in this field, with the goal of improving access to healthcare in underserved areas.

Atsushi Hashimoto from OMRON SINIC X Corporation, will explore the potential of procedural text and video understanding for automating complex tasks. His research has implications for a wide range of applications, from scientific experiments to product assembly.

Amit Gupta from VueMotion, will present a groundbreaking talk on human movement analysis using AI-driven video technology. His expertise in computer vision and machine learning will offer valuable insights into how VueMotion's platform is revolutionizing sports performance and healthcare.

Ryota Hinami from Mantra Inc., will discuss the advancements in manga translation using AI. Learn how LLMs are revolutionizing the industry and addressing challenges in automating this complex process. This talk is a must-attend for those interested in the intersection of AI and language technology.

These talks collectively offer a rich and diverse exploration of the cutting-edge developments in generative AI and multimedia technology. They are sure to inspire and inform attendees from both academia and industry, fostering collaboration and innovation in this exciting field.

5 Industry Demonstrations

We accepted ten intriguing demonstrations showcasing their unique features as follows:

(1) **GaussDB-AISQL**: This innovative system allows users to run complex SQL queries directly over large-scale multimedia data using familiar SQL syntax, eliminating the need for complex Python scripts and offering distributed parallel processing for faster analysis. (2) **RLLTE**: An open-source framework for RL research and application development, providing a complete ecosystem with modular components, extensive benchmarks, and an LLM-powered co-pilot. (3) **vLive**: A cloud-based platform for creating professional-grade virtual live streams with minimal equipment, offering an intuitive user interface with AI-powered features and a vast digital asset library. (4) **DriveCoach**: A smart driving assistance framework that predicts various near-miss risks using multimodal data, offers real-time alerts and personalized behavior recommendations, and provides post-drive coaching for skill improvement. (5) **Canvas**: A framework for interactive art image synthesis that allows artists to integrate customized subject concepts with artistic styles, leveraging a two-stage diffusion-based approach and a large-scale dataset. (6) **Tai-Unbox Creator**: A multi-channel unboxing content generator for Taiwan that creates culturally relevant marketing content by analyzing social media data and understanding local preferences. (7) **Entertainment in an Autonomous World**: A cryptographic framework for fully on-chain gaming, ensuring fairness, privacy, and transparency. The Crystal Cave game demonstrates this framework in action, showcasing a provably fair and secure gaming experience. (8) **MotionBank**: A large motion model for text-driven human motion synthesis that leverages a large-scale dataset of video actions and a rule-based motion captioning algorithm to generate various human motions. (9) **Real-time Interactive Digital Avatar with Efficient Inference**: This work addresses the challenge of creating real-time interactive digital avatars with efficient processing requirements using pre-processed facial animation data and the International Phonetic Alphabet (IPA). (10) **BlitzMe**: A mobile social media app that uses on-device smile recognition to replace the traditional "like" button, offering insights into user engagement and digital well-being.

These demonstrations showcase cutting-edge advancements in multimedia research, encompassing AI-powered content creation, intelligent user assistance systems, and secure and private online experiences. Through these demonstrations the conference fosters innovation and collaboration in multimedia technologies.